This, of course, is a minute subset of the possible linear operators that can be constructed, but we will soon see that **X** and **D** play a very central role in the quantum mechanics of particles.

Now, let's consider the property of Hermiticity. A convenient way to define a Hermitian operator is through its matrix elements, by sandwiching it between a bra and a ket. You can sandwich an operator **L** in two different ways:

$$\langle \Psi | \mathbf{L} | \Phi \rangle$$

or

$$\langle \Phi | \mathbf{L} | \Psi \rangle.$$

In general, there is no simple relation between these two sandwiches. But in the case of a Hermitian operator (for which, by definition, $\mathbf{L}^\dagger = \mathbf{L}$) there is a simple relation: the two sandwiches are complex conjugates of each other:

$$\langle \Psi | \mathbf{L} | \Phi \rangle = \langle \Phi | \mathbf{L} | \Psi \rangle^*.$$

Let's see whether the operators **X** and **D** are Hermitian. Recalling that

$$\mathbf{X}\ \psi(x)\ =\ x\psi(x),$$

and using the inner product formula Eq. 8.2, we can write

$$\langle \Psi | \mathbf{X} | \Phi \rangle = \int\ \psi^*(x) x \phi(x)\, dx$$

and